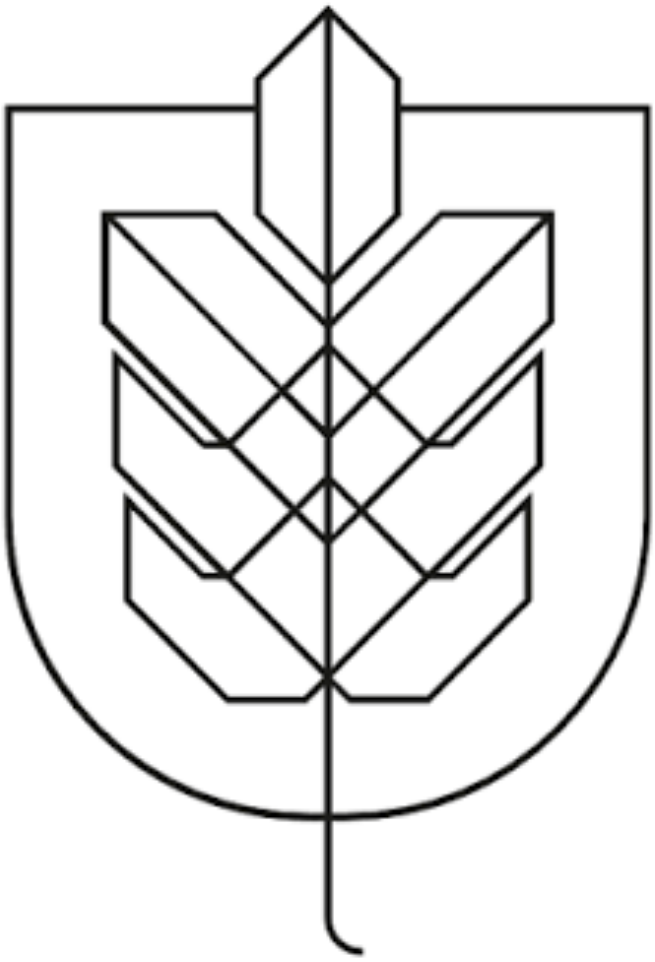


Data Visualization - QTM-6032 - LMMKT1

A2 : Individual Assignment
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2. Executive Summary

The period from 2010 to 2019 showed continuous sales growth which achieved a total of £1,031,439.90 because of strong business operations. The company experienced a significant revenue drop during the first quarter of 2020 which recorded sales at £226,545.90 and brought in only £1,666.60 during April to reach a total of £228,212.50 from January to April. The COVID-19 pandemic created a severe market shock which affected the company's normal Q4 seasonal pattern. The combination of descriptive analytics together with scenario forecasts shows how the economy has contracted and identifies possible recovery solutions which extend until December 2020. The report recommends strict cost and cash controls, accelerated digital transformation, and adaptive forecasting models to sustain revenue, protect margins, and position the business for post-pandemic recovery.

3. Descriptive Analysis

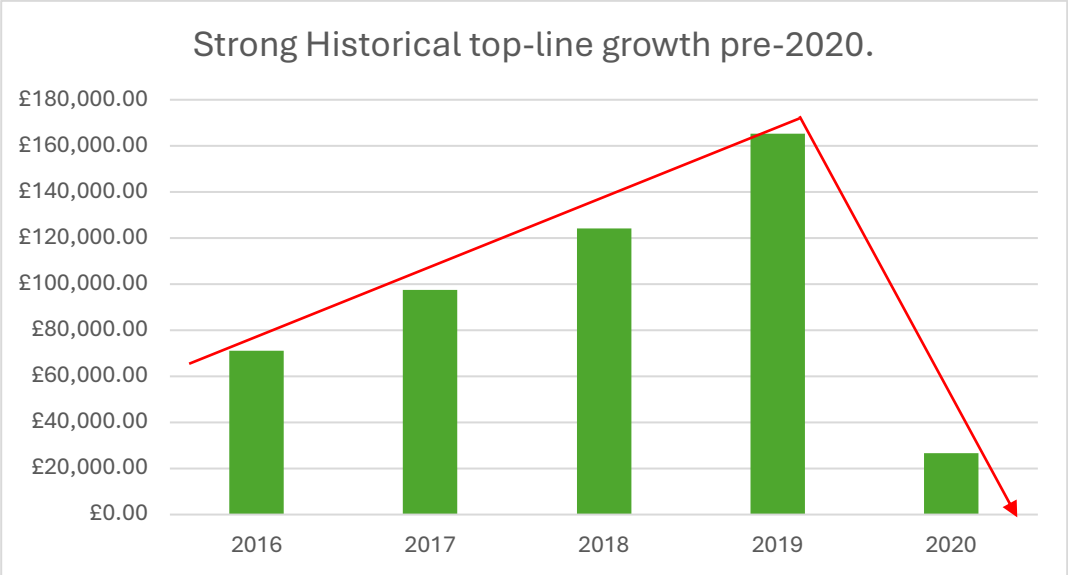
3.1 Data and Scope

This analysis uses monthly sales and profit data from January 2016 to April 2020. We have complete data for the years 2016 to 2019. The data from January to April 2020 is partial, and April shows a very low sales number because of the economic disruption caused by COVID-19. Total sales from January to April 2020 were £228,212.50. In contrast, total sales for the full year in 2019 were £1,031,439.90, which also produced profits of £165,371.60. This dataset allows us to assess growth trends over the long term before 2020 and to evaluate the early disruption in 2020.

3.2 Long-Run Growth (2016 to 2019)

Profit growth 2016 to 2019: profit rose from £71,000 to £165,000, with a CAGR of around 36%. Profit grew faster than sales historically, showing improved operating leverage before 2020. The quicker growth of profits compared to sales implies better operational efficiency and operating leverage, meaning fixed costs were increasingly covered as volumes increased. This trend suggests that the firm achieved scale efficiencies,

improved cost control, or benefited from favourable changes in product mix before 2020. Overall, the period before the pandemic shows a business in strong financial health, positioned for high growth.



Year-on-Year Percentage Change (2016–2020)

From Year → To Year	Profit Change (£)	YoY % Change
2016 → 2017	97,540.00 − 71,145.40 = **26,394.60**	**+37.11%**
2017 → 2018	124,251.30 − 97,540.00 = **26,711.30**	**+27.39%**
2018 → 2019	165,371.60 − 124,251.30 = **41,120.30**	**+33.09%**
2019 → 2020	26,762.70 − 165,371.60 = **−138,608.90**	**−83.82%**

- 2020 shows a sharp drop, but this is because the data covers only January–April.
- Adjusting for a full year gives a more realistic decline of about −51%.

3.3 Central Tendency and Dispersion

In 2019, average monthly sales were about £85,953, close to the median of £86,109, indicating a balanced distribution of monthly revenues. However, the standard deviation was about £25,000 and the coefficient of variation (CV) was 29%, showing significant fluctuations around the mean. This indicates that while the firm had consistent overall growth, seasonal factors and promotional cycles significantly affected monthly performance. In contrast, the early 2020 period (January to April) displayed much greater volatility. Average monthly sales dropped to about £57,000, while the standard

deviation rose sharply to around £43,000, resulting in a coefficient of variation of 76%. This shows extreme instability and reflects the economic shock caused by COVID-19. The sharp increase in fluctuations highlights the collapse of predictable demand patterns, creating significant uncertainty for forecasting, inventory planning, and cash flow management. The comparison between 2019 and early 2020 emphasizes the structural break caused by external economic disruption.

Central Tendencies (2019)	Values
Mean monthly sales	£85,953
Median monthly sales	£86,109
Std dev	£25,000
Coefficient of Variance	29%

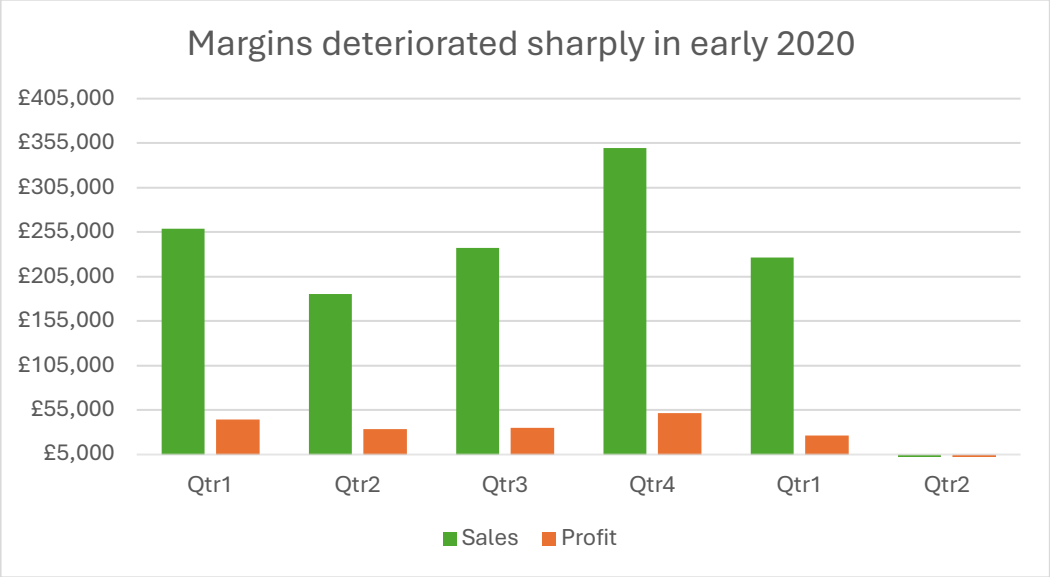
3.4 Seasonality and Sales Concentration

A quarterly analysis of 2019 sales show clear seasonality. The fourth quarter accounted for 34% of annual revenue, followed by the first quarter at 25%, the third quarter at 23%, and the second quarter at just 18%. This heavy concentration in the final quarter shows a strong reliance on peak seasonal demand, likely driven by year-end holidays, promotional campaigns, and gifting behaviour. Such concentration increases business risk. A disruption during the fourth quarter would severely impact annual performance since nearly one-third of revenue is generated in that period. On the other hand, this also represents a strategic chance: effective planning, marketing investment, and supply chain readiness before the fourth quarter could lead to significant returns. The observed seasonality requires effective demand forecasting, flexible inventory allocation, and marketing optimization to take advantage of peak periods and reduce risks.

Year (2019 & 2020)	Sales	Profit
Qtr1	£259,018	£44,697
Qtr2	£185,584	£34,008
Qtr3	£237,515	£34,893
Qtr4	£349,322	£51,773
Qtr1	£226,546	£26,305
Qtr2	£1,666.60	£458.10

3.5 Margin Behaviour

Profitability trends show early signs of operational strain in 2020. In 2019, the firm achieved an overall profit margin of about 16.0%, reflecting strong operational health. However, during the first quarter of 2020 (January to March), the margin fell to approximately 11.6%, with February showing a notably low margin of 4.3%. This sharp drop may reflect a mix of pricing pressures, increased fulfilment and logistics costs, shifts towards lower-margin products, or discounting strategies used to maintain demand. The significant decrease suggests that while the firm kept some sales momentum, it did so at the cost of profitability. This trend highlights the importance of maintaining margin discipline, especially during volatile times, and stresses the need for careful cost control, refining pricing strategies, and optimizing the product portfolio.



3.6 Customer Segment Snapshot (2020 YTD)

Customer segment analysis for January to April 2020 shows significant differences in revenue contribution and profitability. The Consumer segment generated the highest sales at £108,130.80 but only produced profits of £10,483.30, indicating relatively thin margins. Corporate customers contributed £69,507.10 in sales but only yielded £2,541.00 in profits, showing particularly low profitability. In contrast, the Home Office segment generated £50,574.60 in sales while delivering £13,738.40 in profits, making it the most profitable segment based on margin. This suggests that demand from remote workers, driven by the sudden shift to home-based work during COVID-19, has higher profitability potential. Strategically, this segment represents a valuable growth

opportunity, warranting increased marketing focus, tailored product bundles, and targeted customer acquisition efforts.



Home Office Segment Drives Peak Profitability Despite Lower Sales Volume

4. Forecasting & Outlook

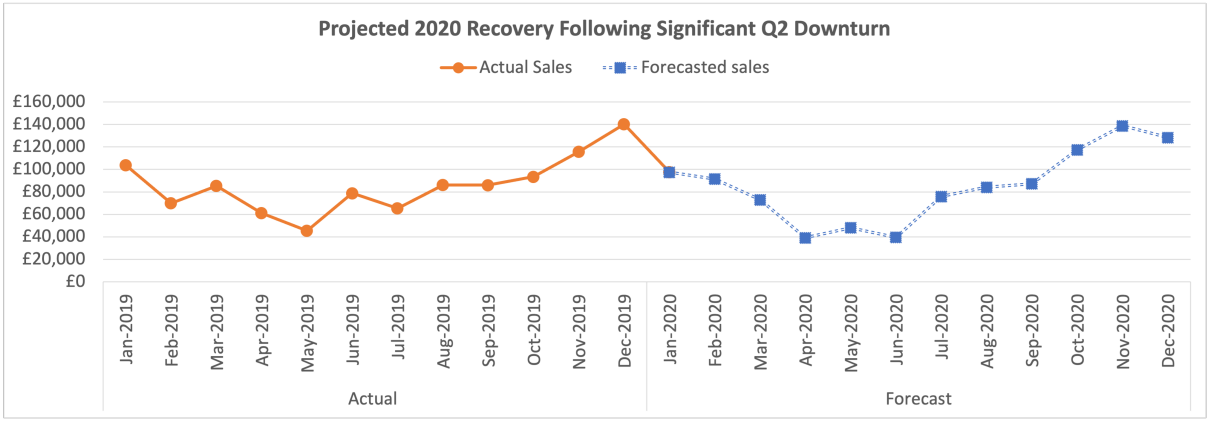
4.1 Forecasting Methodology

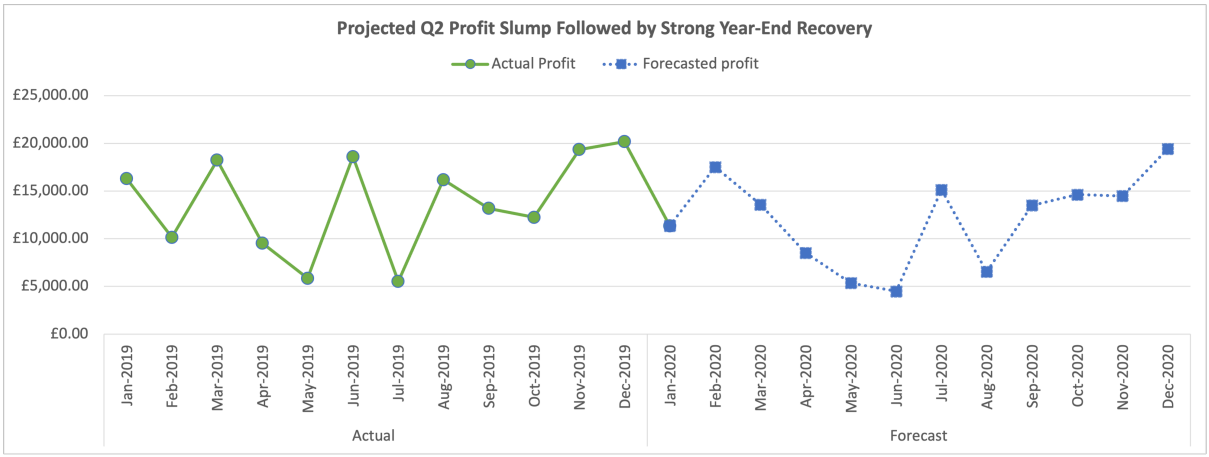
The forecasting method uses a seasonal statistical model which is adjusted through scenario testing to account for the extensive disruptions caused by COVID-19. The researchers established a "normal" seasonal pattern through historical monthly sales data from 2016 to 2019 which showed consistent Q4 peaks and weaker Q2 performance. The baseline was created through Excel's FORECAST.ETS function which predicts future results by analysing common seasonal patterns that occur through a complete business cycle. The baseline forecast required adjustment through scenario multipliers because the ETS algorithm could not detect the structural break which resulted from the pandemic. The observed decline in early 2020 which included a 12.5 percent year-on-year Q1 sales drop and the severe economic contraction in April 2020 served as the basis for deriving the multipliers. The adjusted forecast combines statistical seasonal patterns with actual shock modelling to create an accurate projection for the remainder of 2020.

4.2 Forecast Results

The baseline ETS model projected a continuation of the historical upward trend with strong recovery toward the end of 2020 in line with typical Q4 seasonality. The forecast displayed a completely different path after the implementation of COVID-related adjustments. The Q2 sales forecast showed a large downward adjustment because the total collapse in April 2020 created a situation which forced the company to operate at

minimum capacity until Q4 when partial normal operations began. The adjusted forecast indicates that full-year 2020 sales are likely to fall between 20% and 40% below 2019 levels depending on the pace of recovery. The profit forecasts display a similar pattern because early months show reduced margins which result from operational inefficiencies and demand fluctuations. The profit projection shows slower sales recovery because cost pressures are still affecting the business and it requires strategies that will increase profit margins. The market will experience a sales decline which will continue until the second half of the year when it begins to recover.





4.3 Business Interpretation

The adjusted forecast needs evaluation based on the circumstances from April 2020, which marked the beginning of COVID-19 economic effects. The three key elements which matter most to this situation are the following.

COVID acceleration of online sales. The pandemic brought about an instant shift in consumer behaviour, which made them use digital channels for their shopping needs. The business faces both challenges and opportunities because it experiences strong Q4 sales patterns while offering various products. Your data shows that home office equipment and technology categories already achieve high profitability, which will

increase because remote work demand will continue for a long time. The late 2020 forecasted recovery matches this pattern, which indicates that improved e-commerce capabilities will lead to faster recovery.

Supply chain disruption risk. The April sales decline results from decreased demand and supply chain limitations. The combination of global logistics bottlenecks and inventory shortages together with extended lead times will decrease sales even when demand begins to recover. The risk exists within the conservative Q2 and Q3 multipliers because it has become part of the calculations. The businesses that establish multiple supplier connections while maintaining inventory reserves for high-margin products and enhancing their forecasting methods will achieve greater success in performance stability.

Demand uncertainty. The 2020 early months exhibited unpredictable patterns which demonstrated that consumer and corporate spending became unstable during times of crisis. The forecasted recovery will take time to complete because demand patterns will remain unpredictable throughout all of 2020. The existing uncertainty requires organizations to develop multiple scenarios instead of depending on one specific forecast. The company will use flexible cost structures together with targeted promotions to build its business across home office buyers who represent its most reliable customer segment.

5. Strategic Priorities and Recommendations

The company enters a phase of increased uncertainty and market fluctuations during the upcoming period which extends from 2020. The business shows strong growth trends according to its historical performance records but the data from early 2020 shows a sudden drop in demand and reduced profit margins and unpredictable revenue streams. The management team needs to focus on three essential areas which include preserving liquid assets and maximizing profit margins and advancing digital operations. The organization should execute three strategic priorities which exist in order of decreasing urgency and which elements of the organization need to execute first.

5.1 Priority 1: Protect Cash and Extend Financial Runway

Why: Forecast scenarios indicate a potential 24–40% revenue decline in 2020, while margins fell sharply in early 2020, increasing liquidity risk.

Actions: Freeze non-essential capital expenditure; renegotiate supplier payment terms; tighten receivables collection; secure short-term credit facilities; convert fixed costs into variable where possible.

Impact: Preserves cash, reduces downside risk, and increases operational flexibility during market uncertainty.

5.2 Priority 2: Prioritise High-Margin Segments and Product Lines

Why: The Home Office segment shows the highest profitability per sale in 2020 YTD, driven by remote-working demand, while other segments deliver lower margins.

Actions: Reallocate marketing spend toward Home Office customers; prioritise inventory for high-margin SKUs; bundle profitable accessories; accelerate fulfilment for priority products.

Impact: Improves overall margin, stabilises profitability, and offsets revenue declines.

5.3 Priority 3: Accelerate Digital and Fulfilment Capabilities

Impact: Improves overall margin, stabilises profitability, and offsets revenue declines
Why: COVID-19 has accelerated online demand and urban market concentration, increasing reliance on digital channels.

Actions: Upgrade eCommerce UX; expand fast-shipping options; deploy targeted digital advertising; develop a B2B portal for corporate customers.

Impact: Enhances conversion rates, customer retention, and long-term growth resilience.

6. Conclusion

The report examined eCommerce sales data from previous years to study performance trends and to forecast future performance and to establish strategic goals during the early 2020 period which experienced uncertainty. The analysis demonstrated that consumer behaviour had reached a peak before the pandemic started after which periodic buying patterns emerged and commercial operations experienced declining profits while their business activities became more unpredictable during the COVID-19 pandemic. The forecast scenarios demonstrate that all projected outcomes face major risks which create an urgent need for the organization to implement immediate solutions. The organization needs to establish cash flow protection as its topmost priority while it should direct its business efforts towards high-value customer groups, and it needs to speed up its progress in online sales and distribution operations. The organization will achieve financial stability and profit consistency through these methods which will enable it to build a foundation for continuous market success in the digital business environment.

7. References

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Candia, B., Coibion, O., Gorodnichenko, Y., & Weber, M. (2022). *The macroeconomic expectations of households and businesses during the COVID-19 pandemic*. **Journal of Monetary Economics, 129**, 1–18. https://doi.org/10.1016/j.jmoneco.2022.01.001

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8. Appendices

Forecasting Sales

Source	Monthly Sales	Actual Sales	Forecasted sales
Actual	Jan-2019	£103,788	
	Feb-2019	£69,936	
	Mar-2019	£85,295	
	Apr-2019	£61,211	
	May-2019	£45,529	
	Jun-2019	£78,845	
	Jul-2019	£65,391	
	Aug-2019	£86,109	
	Sep-2019	£86,015	
	Oct-2019	£93,478	
	Nov-2019	£115,642	
	Dec-2019	£140,202	
Forecast	Jan-2020	£97,589	£97,589
	Feb-2020		£91,551
	Mar-2020		£73,007
	Apr-2020		£39,243
	May-2020		£48,108
	Jun-2020		£39,614
	Jul-2020		£75,933
	Aug-2020		£84,162
	Sep-2020		£87,253

	Oct-2020		£117,545
	Nov-2020		£138,912
	Dec-2020		£128,429

Forecasted Profits

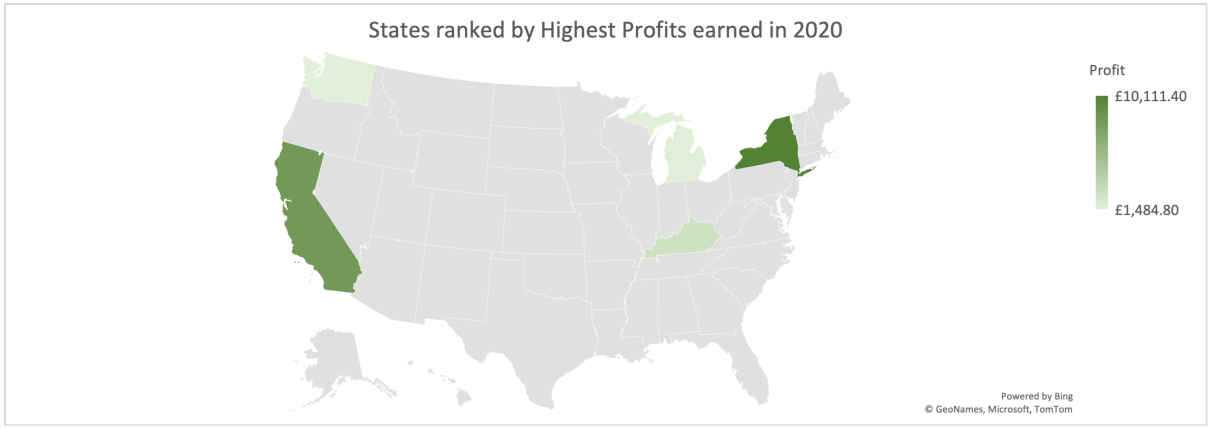
Source	Monthly Profit	Actual Profit	Forecasted profit
Actual	Jan-2019	£16,319.20	
	Feb-2019	£10,140.10	
	Mar-2019	£18,238.10	
	Apr-2019	£9,543.60	
	May-2019	£5,858.80	
	Jun-2019	£18,605.20	
	Jul-2019	£5,518.30	
	Aug-2019	£16,196	
	Sep-2019	£13,179	
	Oct-2019	£12,243	
	Nov-2019	£19,350	
	Dec-2019	£20,180	
Forecast	Jan-2020	£11,346	£11,346
	Feb-2020		£17,528
	Mar-2020		£13,581
	Apr-2020		£8,529
	May-2020		£5,364
	Jun-2020		£4,491
	Jul-2020		£15,142
	Aug-2020		£6,539
	Sep-2020		£13,503
	Oct-2020		£14,619
	Nov-2020		£14,489
	Dec-2020		£19,434

Sales and Profit Segmentation by Geography

Highest profits in states

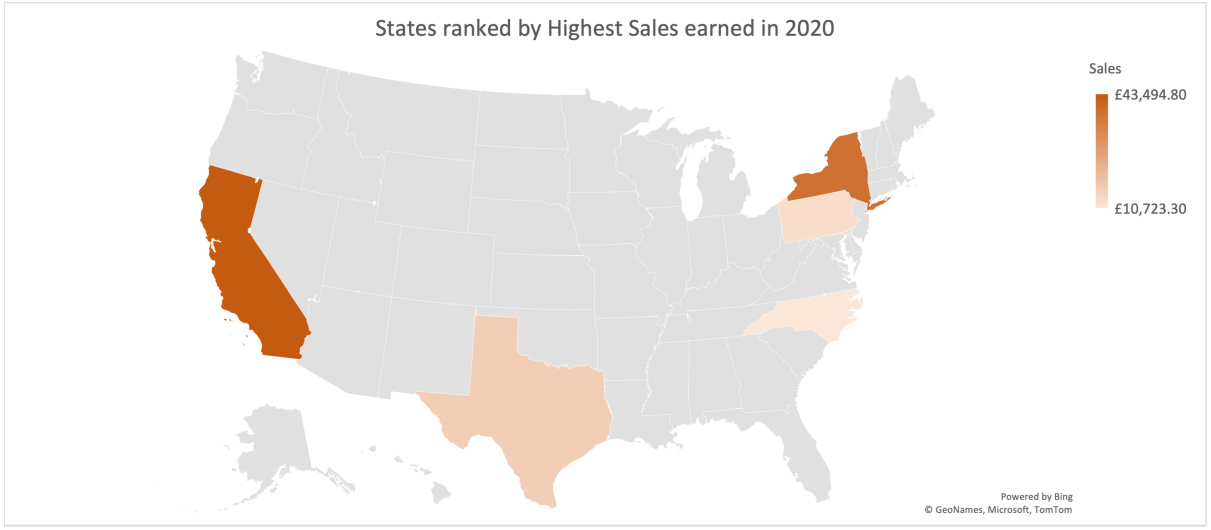
State	Profit	Sales
New York	£10,111.40	£38,347.90
California	£8,288.60	£43,494.80
Kentucky	£2,664.30	£9,227.20
Michigan	£1,527.70	£5,829.90

Washington	£1,484.80	£9,632.60



Highest sales in states

State	Sales	Profit
California	£43,494.80	£8,288.60
New York	£38,347.90	£10,111.40
Texas	£16,060.90	-£3,029.20
Pennsylvania	£12,552.10	-£105.70
North Carolina	£10,723.30	-£3,517.40



Product Segmentation (2020)

Product	Profit	Sales
Furniture	£4,457.00	£62,789.80
Office Supplies	£11,760.20	£94,153.80
Technology	£10,545.50	£71,268.90

Sub products	Sum of Sale Amount
Phones	£27,294.90
Chairs	£26,314.60
Storage	£24,487.50
Binders	£21,186.50
Copiers	£18,131.00
Tables	£14,957.00
Appliances	£14,837.30
Furnishings	£14,804.20
Accessories	£14,130.00
Paper	£14,017.60
Machines	£11,713.00
Art	£8,670.20
Bookcases	£6,714.00
Supplies	£4,366.70
Envelopes	£2,714.10
Labels	£1,986.20
Fasteners	£1,887.70

Highest profits in states for furniture

State	Profits	Sales
Kentucky	£1,112.90	£4,458.60
California	£820.30	£10,806.80
Michigan	£808.50	£3,275.40
New York	£663.70	£4,804.70
Washington	£457.80	£3,564.80

Highest sales in states for furniture

State	Sales	Profits
California	£10,806.80	£820.30
Pennsylvania	£7,387.70	-£320.70
New York	£4,804.70	£663.70
Texas	£4,487.20	-£597.70
Kentucky	£4,458.60	£1,112.90

Highest profits in states for office supplies

State	Profits	Sales
California	£4,322.80	£18,391.20
New York	£3,175.30	£13,004.30
Minnesota	£941.00	£2,480.30
Washington	£934.40	£5,458.10
Kentucky	£837.10	£3,069.10

Highest sales in states for office supplies

State	Sales	Profits
California	£18,391.20	£4,322.80
New York	£13,004.30	£3,175.30
Texas	£8,863.20	-£2,764.20
Washington	£5,458.10	£934.40
Pennsylvania	£4,131.70	£149.30

Highest profits in states for Technology

State	Profit	Sales
New York	£6,272.40	£20,538.90
California	£3,145.50	£14,296.80
Nebraska	£784.00	£2,540.00
Kentucky	£714.30	£1,699.50
Indiana	£582.90	£2,189.40

Highest sales in states for Technology

State	Sales	Profit
New York	£20,538.90	£6,272.40
California	£14,296.80	£3,145.50
North Carolina	£8,771.10	-£3,713.80
Texas	£2,710.50	£332.70
Nebraska	£2,540.00	£784.00

Highest profits in cities for furniture

City	Profit	Sales
Detroit	£808.50	£3,275.40
Louisville	£777.50	£3,222.10
Seattle	£444.30	£3,402.30
New York City	£421.10	£3,662.90
Henderson	£335.40	£1,236.50

Highest sales in cities for furniture

City	Sales	Profit
Philadelphia	£7,387.70	-£320.70
San Francisco	£3,897.30	£172.60
New York City	£3,662.90	£421.10
Seattle	£3,402.30	£444.30
Detroit	£3,275.40	£808.50

Highest profits in cities for office supplies

City	Profit	Sales
New York City	£2,970.90	£11,937.60
San Francisco	£2,030.90	£8,237.00
Los Angeles	£1,123.00	£4,600.40
Minneapolis	£905.90	£2,063.30
Columbia	£720.20	£2,618.80

Highest sales in cities for office supplies

City	Sales	Profit
New York City	£11,937.60	£2,970.90
San Francisco	£8,237.00	£2,030.90
Los Angeles	£4,600.40	£1,123.00
Seattle	£4,025.00	£611.00
Philadelphia	£3,817.70	£199.60

Highest profits in cities for tech

City	Profit	Sales
New York City	£6,108.50	£19,871.50
Los Angeles	£1,598.70	£5,723.30
Richmond	£1,258.50	£3,646.50
San Francisco	£966.20	£4,646.60
Omaha	£784.00	£2,540.00

Highest sales in cities for Technology

City	Sales	Profit
New York City	£6,108.50	£19,871.50
Burlington	-£3,732.40	£8,617.00
Los Angeles	£1,598.70	£5,723.30
San Francisco	£966.20	£4,646.60
Richmond	£1,258.50	£3,646.50

